

SECTION E: Binomial Theorem and Expansion

1. What is the coefficient of x^r in the expansion of $(1 + x)^n$?

A. $\binom{n}{r-1}$

B. $\binom{n+1}{r+1}$

C. $\binom{n}{r} + \dots$

D. $\binom{n-1}{r}$

2. One of the followings has equal value as $\binom{n}{1}$.

A. $\binom{n}{n-1} + \dots$

B. $\binom{n}{r}$

C. $\binom{n}{r}$

D. $\binom{n-1}{r}$

3. From Pascal's triangle, addition of $\binom{n}{r-1}$ and $\binom{n}{r}$ give

A. $\binom{n+1}{r} + \dots$

B. $+\binom{n}{r+1}$

C. $+\binom{n+1}{r-1}$

D. $+\binom{n}{r-1}$

4. In notation form, and considering the expansion in ascending powers of x , $(a + x)^n$ can be expressed as

A. $\sum_{r=1}^n \binom{n}{r} x^r a^{n-r}$

B. $\sum_{r=0}^n \binom{n}{r} x^r a^{n-r} + \dots$

C. $\sum_{r=0}^n \binom{n}{r} x^{n-r} a^{n-r}$

D. $\sum_{r=0}^n \binom{n}{r} x^{n-r} a^r$

5. When we differentiate the expansion of $(1+x)^n$ with respect to x two times, put $x = 0$ and simplify, we get

A. $\binom{n}{r} = \frac{2}{n}$

B. $\binom{n}{1} = \frac{2}{n+1}$

C. $\binom{n}{2} = \frac{n(n-1)}{2} + + +$

D. $\binom{n}{2} = \frac{2}{n(n-1)}$

6. For positive integer n , the factorial of n , denoted by $n!$ is equivalent to

A. $(n-1)!$

B. $n(n+1)(n-2)!$

C. $(n+1)!$

D. $n(n-1)(n-2)! + + +$

7. The value of $\sum_{k=0}^n \binom{n}{k}$ is

A. 2^{n-2}

B. $2^n + + +$

C. $2n$

D. n^2

8. Expand $(x+2y)^7$ in ascending powers of x

A. $x^7 + 14x^6y + 84x^5y^2 + 280x^4y^3 + 560x^3y^4 + 672x^2y^5 + 448xy^6 + 128y^7$

B. $128y^7 + 448xy^6 + 672x^2y^5 + 560x^3y^4 + 280x^4y^3 + 84x^5y^2 + 14x^6y + x^7 + + +$

C. $x^7 + 4x^6y + 84x^5y^2 + 28x^4y^3 + 56x^3y^4 + 672x^2y^5 + 448xy^6 + 128y^7$

D. $128y^7 + 448xy^6 + 62x^2y^5 + 560x^3y^4 + 280x^4y^3 + 8x^5y^2 + 14x^6y + x^7$

9. Find the term independent of y in the expansion of $\left(\frac{x^4}{y^3} + \frac{y^2}{2x}\right)^{10}$

A. $\frac{105}{32}x^{11}$

B. $\frac{15}{32}x^{10}$

C. $\frac{105}{32}x^{10} + \dots$

D. $105x^{10}$

10. Find the maximum coefficient in the expansion of $(3x + 5)^{10}$

A. 26581250

B. 26578250

C. 26578125

D. 265781250 + ...

11. Obtain the first four terms of the expansion of $(1 + 3x)^{\frac{1}{3}}$ in ascending powers of x

A. $1 + x - x^2 + \frac{5}{3}x^3 + \dots$

B. $1 - x - x^2 + \frac{5}{3}x^3 + \dots$

C. $1 - x + x^2 + \frac{5}{3}x^3 + \dots$

D. $1 + x - x^2 - \frac{3}{5}x^3 + \dots$

12. Obtain the expansion of $\sqrt{9 + x^2}$ as far as the term in x^4

A. $3 + \frac{1}{3}x^2 + x - \frac{1}{216}x^4$

B. $3 + \frac{1}{6}x^2 - \frac{1}{216}x^4 + \dots$

C. $3 - \frac{1}{6}x^2 - \frac{1}{216}x^4$

D. $3 + \frac{1}{6}x^2 + \frac{1}{216}x^4$

13. Find the constant term in the expansion of $\left(2z^2 + \frac{1}{z}\right)^9$

A. 670

B. 671

C. 672 + ...

D.673

14. Find the middle term in the expansion of $(x^2 + y^2)^8$.

A. $60xy$

B. $70x^8y^8$ +++

C. $70x^6y^8$

D. $70x^8y^6$

15. Find the fifth term in the expansion of $(-3x - 4y)^6$ in descending order of x .

A. $345x^3y^3$

B. $34560x^3y^3$ +++

C. $34560x^2y^4$

D. $3456x^3y^3$

16. For what value of x is the expansion of $\frac{1}{a+bx}$ valid?

A. $|x| < \frac{a}{b}$ +++

B. $|x| > \frac{a}{b}$

C. $|x| < \frac{1}{b}$

D. $|x| < \frac{a}{2}$

17. Find the sum of the constant coefficients in the expansion of $(3 + 2x)^4$

A. 620

B. 630

C. 625 +++

D. 216

18. Find the first four terms in the expansion of $(1 - 3x^2)^5$ in ascending powers of x

A. $1 - 15x^2 + 90x^4 - 270x^6$ +++

B. $1 + 15x^2 + 90x^4 - 270x^6$

C. $1 - 15x^2 - 90x^4 + 270x^6$

D. $1 + 15x^2 + 90x^4 + 270x^6$

19. State the value of $|x|$ for which the expression $(27 - 6x)^{\frac{-2}{3}}$ can be expanded as a series of ascending powers of x .

A. $|x| < \frac{2}{9}$

B. $|x| < \frac{9}{2} +++$

C. $|x| < \frac{1}{9}$

D. $|x| > \frac{2}{9}$

20. State the condition under which the expansion of $(x + 2y)^{-5}$ will be valid in ascending powers of y .

A. $\left|\frac{y}{x}\right| < \frac{1}{2} +++$

B. $\left|\frac{y}{x}\right| < 1$

C. $\left|\frac{y}{x}\right| > \frac{1}{2}$

D. $\left|\frac{x}{y}\right| < \frac{1}{2}$

21. Simplify $\frac{\binom{n}{r+1}}{\binom{n}{r}}$

A. $\frac{r+1}{n-r}$

B. $n - r$

C. $r + 1$

D. $\frac{n-r}{r+1} +++$

22. Find the term independent of y in the expansion of $\left(\frac{4}{x^5} - \frac{2x}{3y}\right)^{10}$

A. $\frac{1048576}{x^{50}} + \dots$

B. $\frac{1048576}{x^{20}}$

C. $\frac{1048576}{y^{30}}$

D. $\frac{1048576}{x^{30}}$

23. Find the fifth term in the expansion of $(3x + 2y^2)^{12}$ in descending powers of x

A. $5196312x^8y^8$

B. $51963120x^6y^9$

C. $51963120x^7y^8$

D. $51963120x^8y^8 + \dots$

24. Find the value of $\frac{\binom{10}{r}}{\binom{10}{r+1}}$

A. $\frac{r+1}{10-r} + \dots$

B. $\frac{r}{10-r}$

C. $\frac{r-1}{10+r}$

D. $\frac{r}{10+r}$

25. Evaluate the term independent of x in the binomial expansion of $\left(x^2 - \frac{1}{2x}\right)^9$

A. $\frac{16}{21}$

B. $\frac{21}{16} + \dots$

C. 21

D. 16

26. Find the constant term in the expansion of $\left(x^2 - \frac{2}{x}\right)^6$

A. 360

B. 240 +++

C. 140

D. 420

27. Find the value of $\sum_{k=1}^5 k \binom{5}{k}$

A. 60

B. 6

C. 32

D. 80 +++

28. Find the value of $\sum_{r=3}^8 r! \binom{8}{r}$

A. 10802

B. 10956

C. 109536 +++

D. 10804

29. Find the coefficient of x^4 in the expansion of $\left(2x - \frac{3}{x^2}\right)^4$

A. 16 +++

B. 32

C. 42.

D. 14

30. Determine the greatest coefficient in the expansion of $(3x + 1)^8$

A. 17496

B. 20412 +++

C. 2941

D. 204120

31. Find the constant term in the expansion of $(3x + 1)^8$

- A. 1 +++
- B. 2
- C. 3
- D. 4

32. Find the value of x for which the expansion of $\left(2 + \frac{1}{4}x\right)^5$ is valid

- A. $|x| < 2$ +++
- B. $|x| < 4$
- C. $|x| < 8$
- D. $|x| < \frac{1}{2}$

33. Evaluate $\sum_{r=1}^5 (r + 3)!$

- A. 46224 +++
- B. 4624
- C. 4224
- D. 46228

34. Simplify $\binom{n+1}{n-1} + \binom{n}{n-1}$

- A. $\frac{2}{n+3}$
- B. $\frac{2}{n^2+3}$
- C. $\frac{n^2+3n}{2}$ +++
- D. $\frac{n^2-3n}{2}$

35. Find the constant term in the expansion of $\left(\frac{1}{\beta^2} - \beta\right)^{18}$

- A. 18564 +++

B. 1856

C. 64531

D. 185640

36. Simplify $\binom{n+1}{n-1} - \binom{n}{n-1}$

A. $\frac{n^2-3n}{2}$

B. $\frac{n^2+3n}{2}$

C. $\frac{n(n+1)}{2}$

D. $\frac{n(n-1)}{2} + \dots$

37. Simplify $\frac{\binom{n+1}{n-1}}{\binom{n}{n-1}}$

A. $\frac{n-1}{2}$

B. $\frac{n+1}{2}$

C. $\frac{n}{2}$

D. $\frac{n}{2} + 1 + \dots$

38. Find the term independent of y in the expansion of $\left(\frac{x^4}{3y^3} + \frac{y^2}{2x}\right)^5$

A. $\frac{5}{36}x^5 + \dots$

B. $\frac{5}{6}x^5$

C. $\frac{36}{5}x^5$

D. $36x^5$

39. Find the coefficient of x^{14} in the expansion of $(20x^2 - 1)^8$

A. 10240000000

B. 10240000

C. -10240000000 +++

D. 1024000000

40. Take away the coefficient of x^{14} in the expansion of $(20x^2 - 1)^8$ from the coefficient of x^{14} in the expansion of $(1 + x)^{14}$

A. 10240000001 +++

B. 1024000

C. 10240000

D. 1024000001

41. Express n in factorial form.

A. $n!$

B. $(n - 1)!$

C. $\frac{n!}{(n-1)!}$ +++

D. $\frac{n!}{(n-1)}$

42. Find the binomial expansion of $\left(1 + \frac{3}{4}x\right)^4$ in descending powers of x .

A. $\frac{81}{256}x^4 + \frac{27}{16}x^3 + \frac{27}{8}x^2 + 3x + 1$ +++

B. $1 + 3x + \frac{27}{8}x^2 + \frac{27}{16}x^3 + \frac{81}{256}x^4$

C. $\frac{81}{256}x^4 - \frac{27}{16}x^3 + \frac{27}{8}x^2 - 3x + 1$

D. $\frac{81}{256}x^4 - \frac{27}{16}x^3 - \frac{27}{8}x^2 + 3x - 1$

43. Find the coefficient of x^4 in the expansion of $\left(1 + \frac{3}{4}x\right)^4$

A. $\frac{8}{25}$

B. $\frac{81}{25}$

C. $\frac{81}{256} +++$

D. $\frac{81}{26}$

44. Find the coefficient of x in the expansion of $\left(1 + \frac{3}{4}x\right)^4$

A. 1

B. 2

C. 3 +++

D. 31

45. Find the value of n when $\binom{n}{2} = 120$.

A. -15

B. 15

C. -16

D. 16 +++

46. Express $\binom{n+r}{n-r}$ in factorial form

A. $\frac{(n+r)!}{(2r)!(n-r)!} +++$

B. $\frac{(n+r)!}{(2r)!(n-r+1)!}$

C. $\frac{n!}{2!r!}$

D. $\frac{(n-r)!}{(2r)!(n+r)!}$

47. Find the value of n when $\binom{n-2}{1} = 5$

A. 5

B. 6

C. 7 +++

D. 8

48. Express $\frac{n(n-1)}{2}$ in factorial form

A. $\frac{n(n-1)!}{2!}$

B. $\frac{n(n+1)!}{2!}$

C. $\frac{n!}{2!}$

D. $\frac{n!}{(n-2)!2!} + + +$

49. Find the term independent of y in the expansion of $\left(\frac{1}{2}x + 3y\right)^{1000}$

A. $\left(\frac{x}{2}\right)^{1000} + + +$

B. $\frac{x}{2^{1000}}$

C. $\left(\frac{x}{2}\right)^{100}$

D. $\frac{x}{2^{100}}$

50. In the expansion of $\left(\frac{1}{2}x + 3y\right)^{1000}$, find the coefficient of x in the term independent of y .

A. $\frac{1}{2^{100}}$

B. $\frac{1}{2^{1000}} + + +$

C. $\frac{x}{2^{100}}$

D. $\frac{x}{2^{1000}}$